

Performance Analysis of Adaptive Channel Equalization Techniques for QPSK Communication

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Abstract—Wireless communication systems are affected by multipath fading and channel noise, which degrade the received signal and increase the bit error rate (BER). This work evaluates the performance of four channel equalization techniques—One-Tap, Least Mean Squares (LMS), Normalized Least Mean Squares (NLMS), and Recursive Least Squares (RLS)—for a QPSK communication system.

Among the evaluated methods, the RLS equalizer achieves the lowest BER and the fastest convergence. At an SNR of 4 dB, it attains BER values of 0.0114 for SPS = 16 and 0.0011 for SPS = 32, outperforming both LMS and NLMS. While NLMS provides a good trade-off between computational complexity and error performance, LMS has lower computational requirements but converges more slowly, resulting in higher BER.

The results also show that synchronization combined with channel equalization improves receiver performance. Accurate synchronization enhances constellation recovery, reducing BER and improving communication reliability, particularly at moderate SNR values and higher SPS configurations.

Index Terms—QPSK, Channel Equalization, LMS, NLMS, RLS, BER, SNR, One-Tap, Adaptive Filtering

I. INTRODUCTION

The growing demand for reliable long-distance data transmission has led to significant advances in digital communication systems. Wireless communication plays a central role in modern applications by enabling wide-area connectivity, high data rates, and efficient information exchange. However, transmitted signals are affected by channel impairments such as multipath fading, frequency offsets, inter-symbol interference (ISI), and additive noise. These impairments degrade signal quality, increase the bit error rate (BER), and reduce overall system performance [2].

Adaptive channel equalization is commonly used to compensate for these impairments and improve signal recovery at the receiver. The Least Mean Squares (LMS) algorithm is widely adopted because of its simple implementation and low computational complexity. However, its relatively slow convergence can limit performance in time-varying channel

conditions [3]. To address this limitation, several adaptive equalization techniques have been developed to achieve faster convergence, lower steady-state error, and improved robustness under different channel conditions [5], [8], [9], [6].

Recent research has further focused on adaptive equalization for high-speed communication systems, hardware-efficient FPGA implementations, and optimization-based equalizers to improve BER performance while reducing implementation complexity [10], [7], [11]. Synchronization and equalization work together to achieve reliable signal recovery [4], [1].

In this work, a complete Quadrature Phase Shift Keying (QPSK)-based transmission and receiver model is implemented and analyzed. The proposed communication chain integrates pulse shaping, synchronization, and adaptive equalization using One-Tap, LMS, NLMS, and RLS algorithms. System performance is evaluated under different samples-per-symbol (SPS) values and signal-to-noise ratio (SNR) conditions using BER as the primary performance metric. In addition to software simulations, the proposed system is validated through real-time transmission using ADALM-PLUTO Software Defined Radio (SDR), demonstrating the practical effectiveness of the implemented equalization techniques.

II. THE SYSTEM MODEL

The proposed system is a complete QPSK-based digital communication chain which consists a transmitter, a wireless-channel, and a receiver. The model incorporates pulse shaping, matched filter, multiple synchronization steps, and equalization for reliable data transmission over a noisy wireless channel.

A. Transmitter

At the transmitter side, the text message is first converted into a binary sequence. The bits are grouped into pairs and mapped to QPSK symbols. The mapping can be expressed as

$$s_k \in \left\{ \frac{1}{\sqrt{2}}(\pm 1 \pm j) \right\} \quad (1)$$

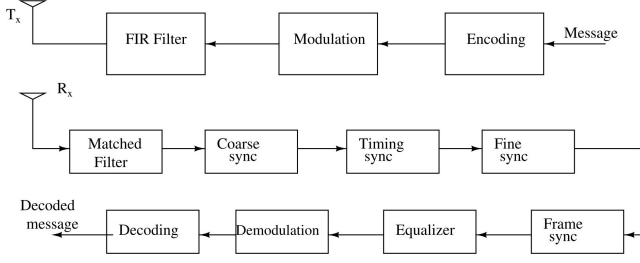


Fig. 1. Overall transmitter and receiver architecture of the proposed system.

where each symbol carries two bits of information.

The symbols are then upsampled by a factor of L (samples per symbol), followed by pulse shaping using a Root Raised Cosine (RRC) filter to limit bandwidth and reduce inter-symbol interference. The transmitted signal is given by

$$x(t) = \sum_k s_k g(t - kT) \quad (2)$$

where $g(t)$ is the RRC pulse and T is the symbol duration.

B. Channel Model

The transmitted signal passes through a wireless channel affected by noise and multipath fading. The received signal is modeled as

$$r(t) = x(t) * h(t) + n(t) \quad (3)$$

where $h(t)$ represents the channel impulse response and $n(t)$ is Additive White Gaussian Noise (AWGN). The multipath channel introduces intersymbol interference, which must be mitigated at the receiver.

C. Receiver

At the receiver, the signal undergoes several processing stages as shown in Fig. 1.

The receiver performs coarse frequency synchronization, timing recovery, Costas loop-based carrier synchronization, frame synchronization, and adaptive equalization before symbol detection.

After synchronization, adaptive equalization is performed to mitigate channel distortion. The architecture of the adaptive equalizer is shown in Fig. 3.

Output of the equalizer $\hat{s}(i)$ and the desired signal $d(i)$ are compared to compute the error

$$e(i) = d(i) - \hat{s}(i) \quad (4)$$

which is used to update the filter coefficients using LMS, NLMS, or RLS algorithms. During training, $d(i)$ is obtained from the known preamble, while in decision-directed mode, it is derived from detected symbols.

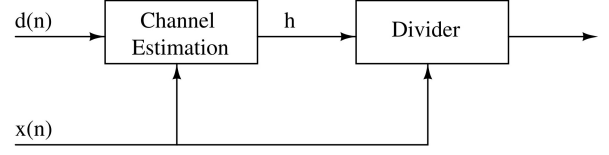


Fig. 2. Channel estimation structure.

III. EQUALIZATION TECHNIQUES

In digital communication systems, transmitted signals are affected by channel impairments such as frequency-selective fading, multipath fading and inter-symbol interference (ISI). These impairments distort the transmitted symbols and increase the bit error rate (BER). Channel equalization compensates for these distortions by estimating the channel response and applying an inverse filter to recover the transmitted symbols. Adaptive equalization techniques have gained significant attention because of their ability to track time-varying wireless channels with low implementation complexity. Recent studies have proposed several improvements to conventional adaptive algorithms for achieving faster convergence and improved BER performance in practical communication systems [5], [6], [10].

In this work, equalization techniques are implemented and compared:

A. One-Tap Equalizer

It is the simplest, suitable for flat-fading channels where the channel can be represented by a single complex coefficient. It estimates this coefficient using a known training sequence and compensates for both amplitude and phase distortions, as illustrated in Fig. 2.

The equalized signal is obtained as

$$y(n) = \frac{x(n)}{h} \quad (5)$$

where $x(n)$ is the received signal and h is the estimated channel coefficient.

The channel estimate is obtained using a known training (preamble) sequence as

$$h = \frac{\sum x(n)d^*(n)}{\sum |d(n)|^2} \quad (6)$$

where $d(n)$ denotes the known transmitted symbols and $(\cdot)^*$ represents the complex conjugate.

B. Adaptive Equalizer

Unlike fixed equalizers, adaptive equalizers continuously update their filter coefficients according to channel variations. As shown in Fig. 3, the synchronized received signal $r_f(i)$ is processed to generate the equalizer output $y(i)$. The output is compared with the desired signal $d(i-n)$ to generate the error signal $e(i)$, which is then used to update the filter coefficients iteratively.

Here, $s(i)$ denotes the transmitted signal, $r_f(i)$ is the synchronized received signal, $y(i)$ is the equalizer output,

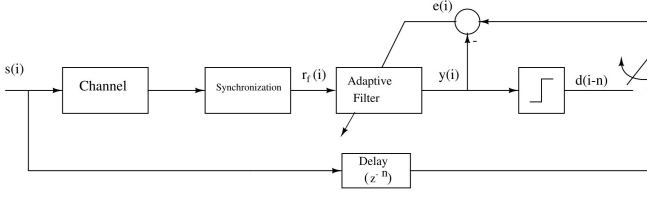


Fig. 3. Adaptive equalizer.

$d(i-n)$ is the delayed desired signal, and $e(i)$ is the estimation error.

C. Least Mean Squares (LMS) Equalizer

The LMS equalizer is one of the most widely used adaptive filtering algorithms because of its simple implementation and low computational complexity. It updates the filter coefficients iteratively by minimizing the mean square error (MSE) between the desired signal and the equalizer output using the method of steepest descent.

The equalizer output is

$$y(n) = \mathbf{w}^H(n)\mathbf{x}(n) \quad (7)$$

The corresponding error signal is

$$e(n) = d(n) - y(n) \quad (8)$$

The coefficients of filter are updated as

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu \mathbf{x}(n)e^*(n) \quad (9)$$

where step size(μ) controlling the convergence speed and stability. A larger value resulting in fast convergence but may lead to instability, whereas a smaller value improves stability at the expense of slower convergence. Consequently, selecting an appropriate step size is critical for achieving reliable equalization performance.

D. Normalized Least Mean Squares (NLMS) Equalizer

The NLMS algorithm improves the LMS algorithm by normalizing the adaptation step with the instantaneous input signal power. This normalization makes the convergence less sensitive to variations in input signal amplitude and improves stability.

The coefficient update equation is

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu}{\|\mathbf{x}(n)\|^2 + \epsilon} \mathbf{x}(n)e^*(n) \quad (10)$$

where ϵ (a small positive constant) is used to prevent division by zero. Compared with LMS, NLMS generally provides faster convergence and improved tracking capability under varying channel conditions while maintaining similar computational complexity.

E. Recursive Least Squares (RLS) Equalizer

The RLS algorithm minimizes the weighted least-squares cost function using both the current and previous input samples. Unlike LMS-based algorithms, RLS exploits past observations through the inverse correlation matrix, resulting in significantly faster convergence and improved BER performance.

The gain vector is computed as

$$\mathbf{k}(n) = \frac{\mathbf{P}(n-1)\mathbf{x}(n)}{\lambda + \mathbf{x}^H(n)\mathbf{P}(n-1)\mathbf{x}(n)} \quad (11)$$

The estimation error is

$$e(n) = d(n) - \mathbf{w}^H(n-1)\mathbf{x}(n) \quad (12)$$

The filter coefficients are updated using

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \mathbf{k}(n)e^*(n) \quad (13)$$

The inverse correlation matrix is updated as

$$\mathbf{P}(n) = \frac{1}{\lambda} [\mathbf{P}(n-1) - \mathbf{k}(n)\mathbf{x}^H(n)\mathbf{P}(n-1)] \quad (14)$$

where λ is the forgetting factor.

F. Detection

After equalization, the recovered symbols are demapped to their corresponding binary bits based on the sign of the real and imaginary components. The overall system performance is evaluated using the BER, defined as

$$\text{BER} = \frac{\text{Number of bit errors}}{\text{Total number of transmitted bits}}. \quad (15)$$

IV. SIMULATION PARAMETERS

The proposed QPSK communication system is analyzed through simulation under controlled conditions. Both SDR-based transmission and software simulations are considered to analyze the effect of synchronization and equalization techniques. The key parameters used in the simulation are described below.

A. System Parameters

The signal is upsampled with different samples-per-symbol (SPS) values to study its impact on performance. A known preamble sequence (11-bit barker code) is used for synchronization and initial training of the equalizers.

- Sampling rate: $f_s = 1$ MHz
- Center frequency: $f_c = 915$ MHz
- Modulation: QPSK
- Samples per symbol (SPS): 4, 8, 16, 32
- RRC roll-off factor: $\beta = 0.35$

B. Channel Parameters

- Channel type: Multipath + AWGN
- SNR Range: -20 dB to $+18$ dB
- Step size: 2 dB

C. Equalizer Parameters

- Equalizer length: $L = 15$
- LMS step size: $\mu = 0.001$ (training), 0.0005 (decision-directed)
- NLMS step size: $\mu = 0.01$ (training), 0.001 (decision-directed)
- Regularization factor: $\epsilon = 10^{-3}$
- RLS forgetting factor: $\lambda = 0.999$
- RLS initialization: $\delta = 10$

V. RESULTS

In results section, the performance of our designed system is evaluated using BER at different SNR conditions for varying SPS values is presented. Both tabular and graphical results are used for analysis.

A. BER Performance for Different SPS

TABLE I
BER PERFORMANCE FOR SPS = 8

SNR (dB)	Before Eq	1-TAP	LMS	NLMS	RLS
-4	0.49840	0.49549	0.49847	0.49861	0.45611
0	0.43968	0.43891	0.43975	0.43961	0.33000
4	0.20299	0.20431	0.20312	0.20319	0.14604
8	0.02799	0.02819	0.02799	0.02799	0.01903

TABLE II
BER PERFORMANCE FOR SPS = 16

SNR (dB)	Before Eq	1-TAP	LMS	NLMS	RLS
-4	0.47500	0.47813	0.47493	0.47486	0.37562
0	0.22389	0.22361	0.22389	0.22375	0.16632
4	0.01236	0.01236	0.01243	0.01243	0.01146
8	0.00000	0.00000	0.00000	0.00000	0.00000

TABLE III
BER PERFORMANCE FOR SPS = 32

SNR (dB)	Before Eq	1-TAP	LMS	NLMS	RLS
-4	0.29444	0.28965	0.29444	0.29458	0.20722
0	0.01882	0.01861	0.01882	0.01882	0.01542
4	0.00111	0.00111	0.00118	0.00111	0.00111
8	0.00000	0.00000	0.00000	0.00000	0.00000

B. BER vs SNR Analysis

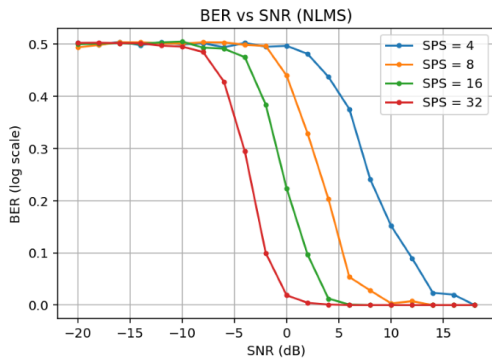


Fig. 4. BER performance of NLMS equalizer for different SPS values.

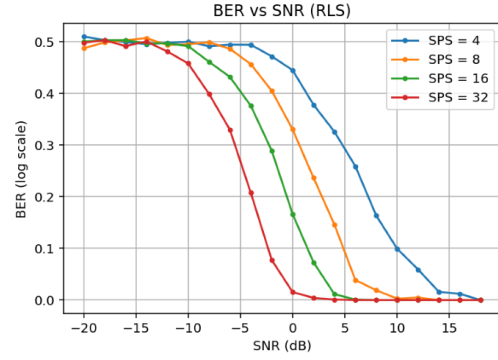


Fig. 5. BER performance of RLS equalizer for different SPS values.

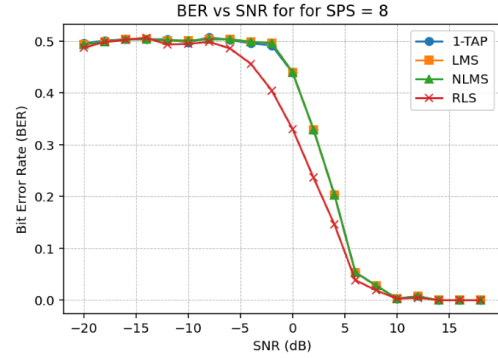


Fig. 6. BER comparison of equalizers for SPS = 8.

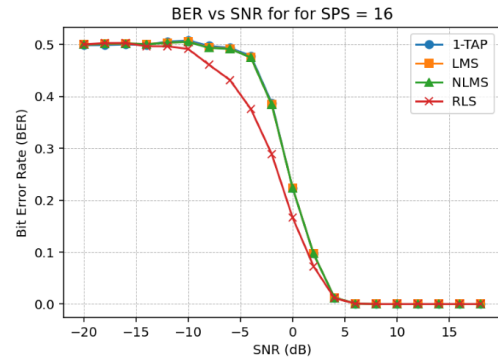


Fig. 7. BER comparison of equalizers for SPS = 16.

Fig. 4 shows that BER decreases with increasing SNR, with SPS = 32 providing the best performance.

As shown in Fig. 5, the RLS equalizer outperforms NLMS in terms of convergence speed and BER reduction. As shown in Fig. 6, for SPS = 8, all equalizers show similar performance at low SNR. As SNR increases, RLS outperforms LMS and NLMS, while the 1-TAP equalizer shows limited improvement.

As illustrated in Fig. 7, at SPS = 16, the performance gap between RLS and other algorithms becomes more significant. From Fig. 8, it is evident that for SPS = 32, all equalizers perform significantly better.

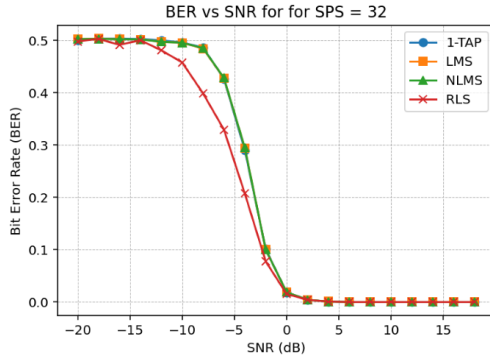


Fig. 8. BER comparison of equalizers for SPS = 32.

C. Observations

From the results, it is observed that:

- BER decreases as SNR increases for all methods.
- Adaptive equalizers significantly improve performance compared to the system without equalization.
- RLS provides the best performance but at higher computational complexity.
- NLMS offers a good balance between the complexity and performance.

VI. DISCUSSION

This section evaluates the performance of the proposed system using both BER measurements and constellation diagrams obtained at different stages of the receiver. The evolution of the constellation illustrates how each synchronization stage and the equalization process progressively improve signal quality.

A. Effect of Synchronization Stages

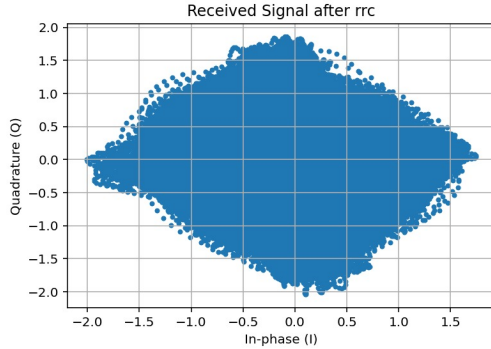


Fig. 9. Signal received after RRC filtering

As shown in Fig. 9, the constellation obtained after RRC filtering is highly distorted, with no well-defined symbol clusters because of channel impairments and synchronization errors.

Following coarse frequency synchronization, the constellation becomes more centered, indicating that the frequency offset has been partially corrected. However, the symbol clusters remain dispersed.

Timing synchronization further improves the alignment of the received symbols, producing more distinct clusters, although residual phase errors are still present.

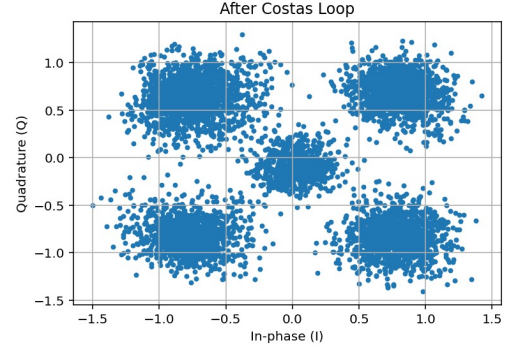


Fig. 10. Costas loop output

As shown in Fig. 12, the Costas loop compensates for the remaining phase and frequency offsets, resulting in constellation points that closely match their ideal locations. This final synchronization stage significantly improves symbol alignment and prepares the signal for accurate detection.

B. Equalization and Performance Analysis

Adaptive equalization mitigates channel distortions, resulting in improved constellation recovery and more accurate symbol detection.

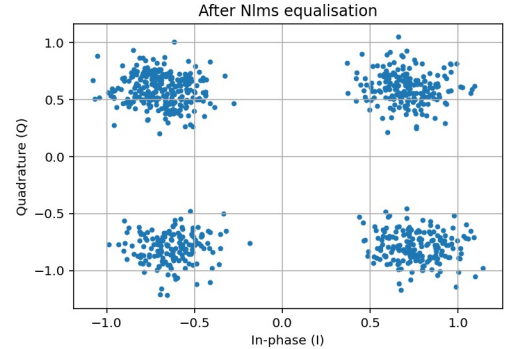


Fig. 11. Constellation after NLMS equalization.

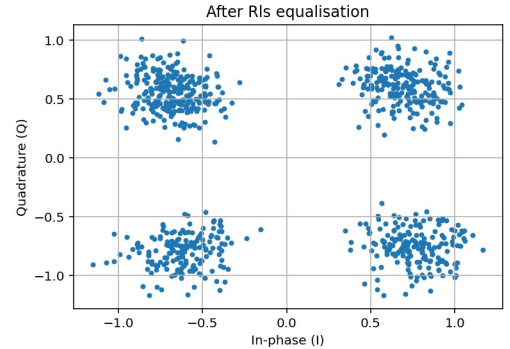


Fig. 12. Constellation after RLS equalization.

Among the evaluated equalizers, RLS achieves the lowest BER and the fastest convergence. NLMS provides a balance between performance and computational complexity, while

LMS offers a simple implementation at the expense of slower convergence.

Increasing the number of samples per symbol (SPS) further improves symbol recovery, leading to lower BER across all evaluated SNR values. The combined use of synchronization and adaptive equalization enables reliable constellation recovery under different operating conditions.

C. Computational Complexity

TABLE IV
COMPUTATIONAL COMPLEXITY OF EQUALIZATION ALGORITHMS

Algorithm	Complexity	Remarks
One-Tap	$O(1)$	Lowest computational cost
LMS	$O(N)$	Simple implementation, slow convergence
NLMS	$O(N)$	Improved stability through normalization
RLS	$O(N^2)$	Fast convergence with higher complexity

The One-Tap equalizer requires only a single channel coefficient estimate, giving it a computational complexity of $O(1)$. LMS and NLMS both have a complexity of $O(N)$, where N denotes the equalizer length. RLS has a complexity of $O(N^2)$ because of the recursive matrix operations involved in coefficient updates. Although RLS is computationally more demanding, it provides the fastest convergence and the lowest BER among the evaluated algorithms.

VII. HARDWARE IMPLEMENTATION

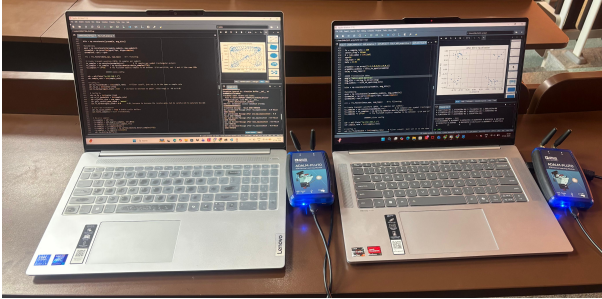


Fig. 13. Experimental setup using ADALM-PLUTO SDR for QPSK transmission and reception.

The proposed communication system was validated using an ADALM-PLUTO Software Defined Radio (SDR) testbed, as shown in Fig. 13. Two laptops were used, with one configured as the transmitter and the other as the receiver. Each laptop was connected to an ADALM-PLUTO SDR module to enable real-time QPSK signal transmission and reception.

The hardware measurements are consistent with the simulation results, confirming reliable system operation under practical conditions.

VIII. CONCLUSION

This work evaluated a QPSK communication system incorporating synchronization and adaptive channel equalization under different channel conditions. The key observations are as follows:

- BER decreases with increasing SNR. For SPS = 32, the BER is approximately 0.018 at 0 dB and approaches zero for SNR values above 8 dB.
- RLS provides the best performance among the evaluated equalizers, achieving the lowest BER (approximately 0.0011 at 4 dB for SPS = 32) together with the fastest convergence.
- Increasing the number of samples per symbol (SPS) improves symbol recovery and reduces BER over the entire SNR range.

The results show that synchronization combined with adaptive equalization enables reliable QPSK communication under moderate SNR conditions. Among the evaluated techniques, RLS provides the best error-rate performance, whereas NLMS offers a lower-complexity alternative with competitive performance. Future work will consider higher-order modulation schemes and more challenging wireless channel models.

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